



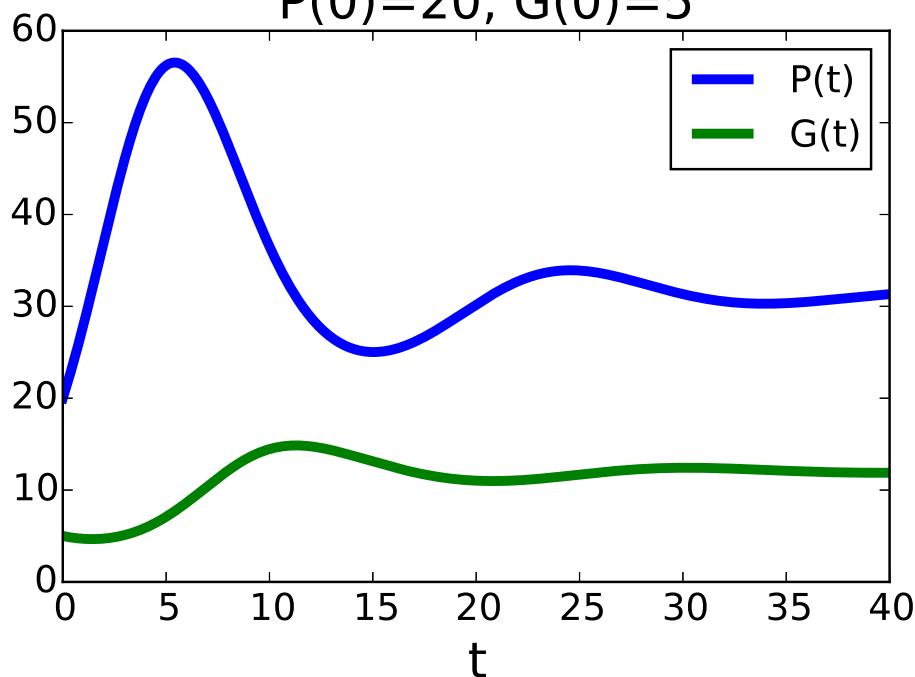
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Calculation (phase plane)

Calculation (phase plane)

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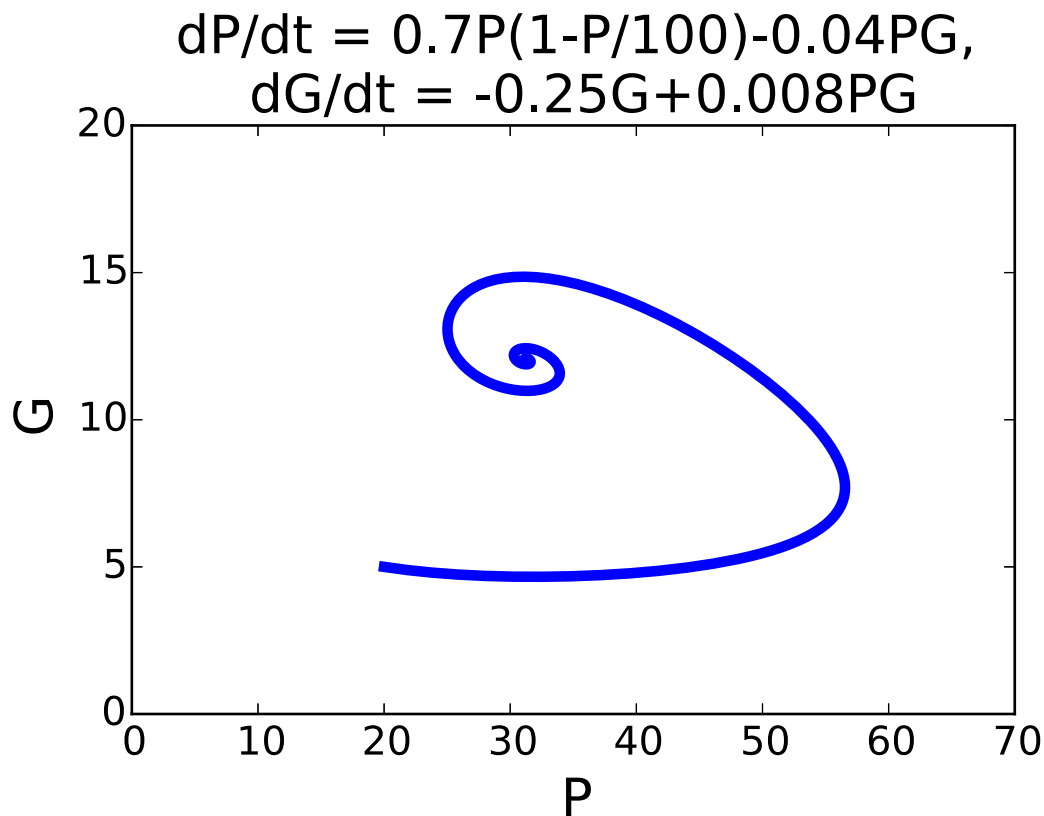
On the previous page, you have learned to approximate solutions of a system of differential equations with Euler's method. It is the same as for single differential equations, only now you work with a vector instead of a scalar. A graph of the solutions of the initial value problem is the following.

$$\begin{aligned} dP/dt &= 0.7P(1-P/100)-0.04PG, \\ dG/dt &= -0.25G+0.008PG, \\ P(0) &= 20, G(0) = 5 \end{aligned}$$

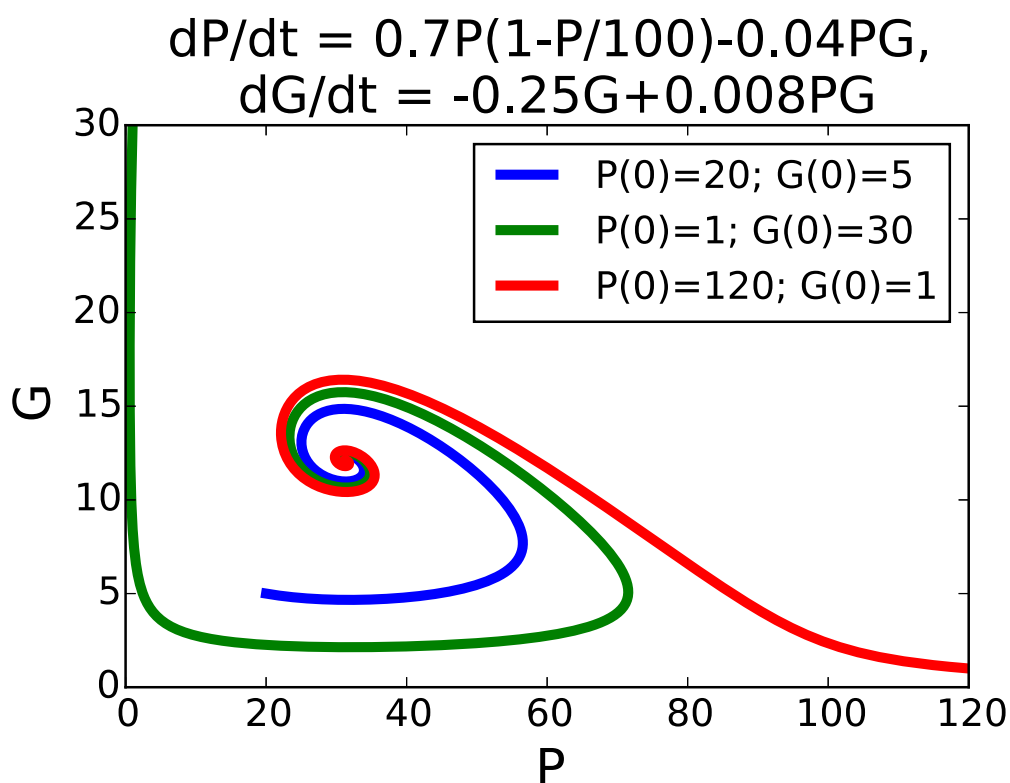


Both P and G have been plotted as functions of time t . Another way to visualise these solutions is to plot P and G against each other: to draw the solutions in the phase plane.

The *phase plane* is in this case the PG -plane. At any time t the solutions P and G constitute a point $(P(t), G(t))$ in the phase plane. The collection of points $(P(t), G(t))$ for parameter t in an interval form a curve in the phase plane.



A solution curve drawn in the phase plane is called a *trajectory*. If you take another initial value for P and/or G , you obtain a different solution, and you can add its trajectory to the phase plane as well. A few trajectories in a phase plane constitute a *phase portrait* of the differential equations. An example of a phase portrait is the following graph:



Exercise A

1/1 point (graded)

Try to recreate the phase portrait above.

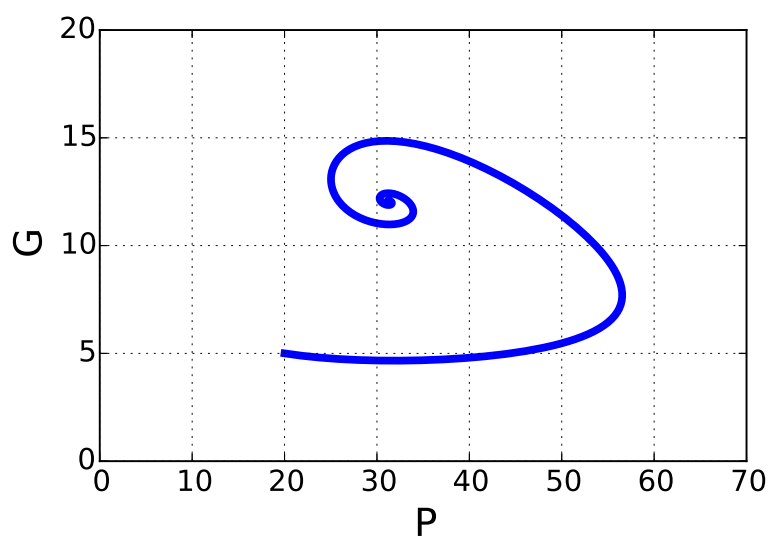
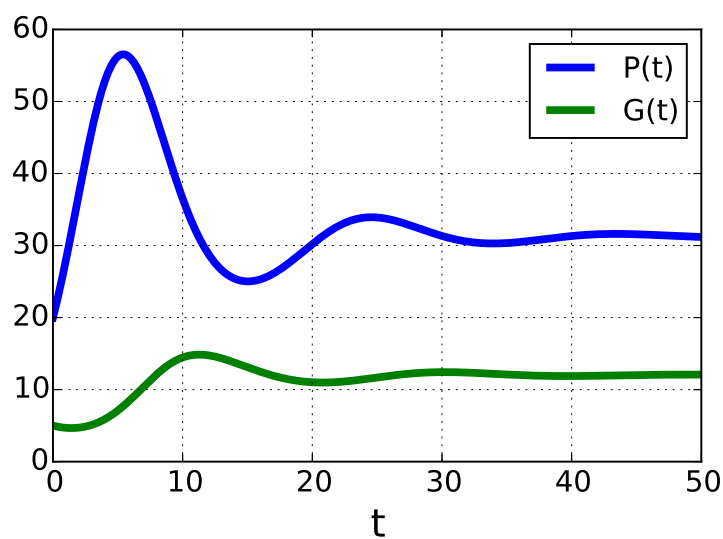
Did you manage to recreate (the important parts) of the figure?

☒ Yes. ✓

☐ No.

Submit

You have used 1 of 10 attempts



You see that after about a month, both populations stabilise: the number of rainbowfish to a little over **30** fish and the gouramis to approximately **12** fish. You can estimate the precise values of the end states with Euler's method. Another way is to calculate the equilibrium points of the system of differential equations.

For a system of differential equations, $\frac{d\vec{X}}{dt} = \vec{F}(t, \vec{X})$, an *equilibrium point* $\vec{X}_0$ is a solution of

$$\frac{d\vec{X}_0}{dt} = \vec{F}(t, \vec{X}_0) = \vec{0}.$$

A different name for equilibrium point is *stationary point*: a point in which the dependent variables stay stationary: they will not change.

So, in our case, the equilibrium points are the points (P, G) where

$$\text{both } \frac{dP}{dt} = 0 \quad \text{and} \quad \frac{dG}{dt} = 0.$$

That implies that:

$$\begin{cases} \frac{dP}{dt} = 0.7 P(t) - 0.007 P^2(t) - 0.04 P(t)G(t) = 0, \\ \frac{dG}{dt} = -0.25 G(t) + 0.008 P(t)G(t) = 0. \end{cases}$$

These equations can be rewritten as:

$$\begin{cases} \frac{dP}{dt} = 0.7 P(t) \left(1 - \frac{P(t)}{100} - \frac{0.04}{0.7} G(t) \right) = 0, \\ \frac{dG}{dt} = -0.25 G(t) \left(1 - \frac{0.008}{0.25} P(t) \right) = 0. \end{cases}$$

Exercise B

1.0/1.0 point (graded)

Use the equations for the equilibrium points and calculate the exact end states.

Give your answers as a fraction, or with 5 decimals.

$$\lim_{t \rightarrow \infty} P(t) =$$



$$\lim_{t \rightarrow \infty} G(t) =$$



You have used 1 of 3 attempts

Exercise C (1 point possible)

Question

How many equilibrium points does the system for the rainbowfish and gouramis have? In your explanation add their coordinates.



Answer



Reflection



Results

Correct Answer

3 equilibrium points:

(0,0), (100,0) and (125/4, 385/32).

Your initial answer: 3 equilibrium points: (Option 3)

👤 "(0,0), (100,0), (31.25000, 12.03125)"

Your final answer: 3 equilibrium points: (Option 3)

👤 "(0,0), (100,0), (31.25000, 12.03125)"

See how your classmates answered below

Class Breakdown

This is a look at how your classmates answered the question during the initial and final rounds.

Answer Options

Option 1

1 equilibrium point:

Option 2

2 equilibrium points:

Option 3 (correct)

3 equilibrium points:

Option 4

4 equilibrium points:

Initial Answer Selection

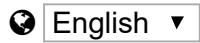
Not enough data to generate the chart. Please check back later.

Final Answer Selection

Not enough data to generate the chart. Please check back later.

On the next page, you will review the results for the aquarium.

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